

CFD Assignment 5

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I. PROBLEM STATEMENT

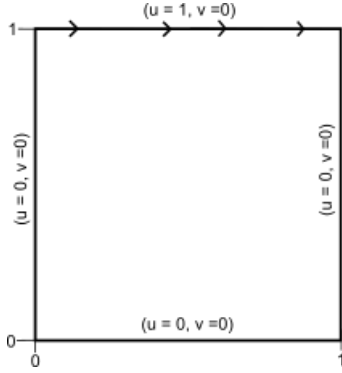


Fig. 1: Schematic of the lid-driven cavity flow problem.

The objective of this project is to numerically simulate the steady-state, two-dimensional, incompressible flow within a square cavity driven by the motion of the top lid, as illustrated in Figure 1. The governing equations are formulated using the vorticity–stream function approach. The velocity field is obtained from the stream function, while the transport of vorticity is governed by the vorticity transport equation. The simulation is conducted for a Reynolds number of 100, and the solution is continued until a steady-state flow field is achieved.

TABLE I: Simulation Parameters

Parameter	Value
Domain Size	1×1
Grid Size	31×31
Reynolds Number	100
Lid Velocity (U_0)	1
Time Step (adaptive)	CFL condition based
Convergence Tolerance	10^{-8}

II. DISCRETIZED EQUATIONS FOR LID-DRIVEN CAVITY PROBLEM

A. Poisson Equation for Stream Function ψ

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\omega$$

Discretized using central differences:

$$\psi_{i,j}^{(n+1)} = \frac{\psi_{i+1,j}^{(n)} + \psi_{i-1,j}^{(n)} + \beta^2 (\psi_{i,j+1}^{(n)} + \psi_{i,j-1}^{(n)})}{2(1 + \beta^2)} + \frac{\Delta x^2 \omega_{i,j}}{2(1 + \beta^2)}, \quad \text{where } \beta = \frac{\Delta x}{\Delta y}$$

B. Velocity Components from Stream Function

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

Discretized using central differences:

$$u_{i,j} = \frac{\psi_{i,j+1} - \psi_{i,j-1}}{2\Delta y}, \quad v_{i,j} = -\frac{\psi_{i+1,j} - \psi_{i-1,j}}{2\Delta x}$$

C. Time Step from CFL Condition

$$\Delta t_c = \sigma_c \frac{\Delta x \Delta y}{|u|_{\max} \Delta y + |v|_{\max} \Delta x}$$

$$\Delta t_d = \sigma_d \frac{1}{2\nu} \frac{\Delta x^2 \Delta y^2}{\Delta x^2 + \Delta y^2}$$

$$\Delta t = \min(\Delta t_c, \Delta t_d)$$

D. Vorticity Transport Equation

$$\frac{\partial \omega}{\partial t} + u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} = \nu \left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right)$$

1) Convection Term (2nd Order Upwind):

$$\left(\frac{\partial \omega}{\partial x} \right)_{i,j} \approx \begin{cases} \frac{3\omega_{i,j} - 4\omega_{i-1,j} + \omega_{i-2,j}}{2\Delta x}, & \text{if } u_{i,j} > 0 \\ \frac{-3\omega_{i,j} + 4\omega_{i+1,j} - \omega_{i+2,j}}{2\Delta x}, & \text{if } u_{i,j} \leq 0 \end{cases}$$

$$\left(\frac{\partial \omega}{\partial y} \right)_{i,j} \approx \begin{cases} \frac{3\omega_{i,j} - 4\omega_{i,j-1} + \omega_{i,j-2}}{2\Delta y}, & \text{if } v_{i,j} > 0 \\ \frac{-3\omega_{i,j} + 4\omega_{i,j+1} - \omega_{i,j+2}}{2\Delta y}, & \text{if } v_{i,j} \leq 0 \end{cases}$$

For points closer to the boundary, 1st order schemes were employed

2) Diffusion Term (Central Difference):

$$\nu \left(\frac{\omega_{i+1,j} - 2\omega_{i,j} + \omega_{i-1,j}}{\Delta x^2} + \frac{\omega_{i,j+1} - 2\omega_{i,j} + \omega_{i,j-1}}{\Delta y^2} \right)$$

3) Time Integration (Explicit Euler):

$$\omega_{i,j}^{n+1} = \omega_{i,j}^n + \Delta t \left[- \left(u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} \right) + \nu \left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right) \right]$$

E. Vorticity Boundary Conditions

$$\begin{aligned} \text{Top (lid): } \omega_{i,N_y} &= -\frac{2(\psi_{i,N_y} - \psi_{i,N_y-1})}{\Delta y^2} - \frac{2U_0}{\Delta y} \\ \text{Bottom: } \omega_{i,1} &= -\frac{2(\psi_{i,2} - \psi_{i,1})}{\Delta y^2} \\ \text{Left: } \omega_{1,j} &= -\frac{2(\psi_{2,j} - \psi_{1,j})}{\Delta x^2} \\ \text{Right: } \omega_{N_x,j} &= -\frac{2(\psi_{N_x-1,j} - \psi_{N_x,j})}{\Delta x^2} \end{aligned}$$

III. NUMERICAL IMPLEMENTATION

The simulation was implemented in Python using NumPy for numerical operations and Matplotlib for visualization and animation. The code solves the lid-driven cavity flow using the stream function-vorticity formulation. The following points summarize the key implementation details:

- **Domain and Grid:** The domain is a unit square cavity ($L = H = 1$), discretized using a uniform grid of 31×31 points in both x and y directions.
- **Governing Parameters:** Reynolds number $Re = 100$, lid velocity $U_0 = 1.0$, and the kinematic viscosity is calculated as $\nu = \frac{U_0 L}{Re}$.
- **Time Stepping:** The time step Δt is dynamically computed using both convection ($\sigma_c = 0.4$) and diffusion ($\sigma_d = 0.6$) CFL conditions to ensure stability.
- **Boundary Conditions:**
 - Top lid moves with velocity U_0 , imposing shear and stream function boundary conditions.
 - All other walls are stationary and modeled using appropriate conditions.
- **Solvers:**
 - The stream function Poisson equation is solved iteratively.
 - Velocity fields (u, v) are obtained from the stream function using central differences.
 - Vorticity transport is solved explicitly with an upwind-biased scheme for the convective term and second-order central differences for diffusion.
- **Convergence:** The simulation iterates until the RMS error in velocity fields drops below 10^{-8} or the maximum iteration count (50,000) is reached.
- **Validation:** Results were compared against benchmark data from Ghia et al. **ghia1982** by plotting u and v velocities along the vertical and horizontal centerlines, respectively.
- **Post-Processing:** Visualization includes:
 - Contour plots of stream function ψ .
 - Streamlines generated using ψ and velocity fields.
 - Centerline velocity comparisons with literature.
 - Log plot of RMS error history to track convergence.
 - An animation showing evolution of ψ and streamlines over time.

The full Python code is submitted along with this report.

IV. RESULTS AND DISCUSSION

A. Convergence Behavior

The convergence history of the simulation is illustrated in Figure 2, where the logarithm of the RMS error of u and v is plotted against the number of iterations. The plot demonstrates a consistent and monotonic decay in RMS error of both u and v , indicating stable convergence of the numerical scheme. The simulation was terminated when the error fell below a predefined tolerance of 10^{-8} , ensuring that the solution had reached a quasi-steady state.

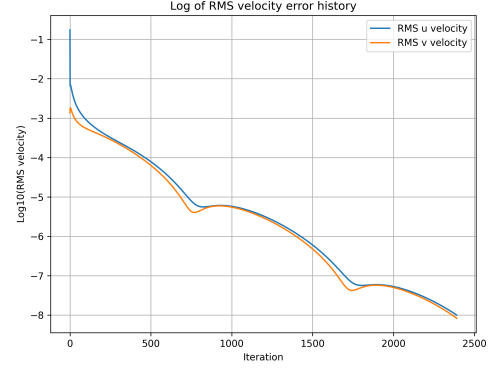


Fig. 2: Logarithmic error history of the simulation.

B. Stream Function Contours

The stream function contours presented in Figure 3 reveal the internal flow structure of the cavity. The primary vortex dominates the center of the cavity, slightly skewed to the right due to the motion of the lid. Additionally, smaller secondary vortices appear in the bottom corners, capturing the recirculating flow near the boundaries. These features are consistent with physical expectations and previous benchmark studies.

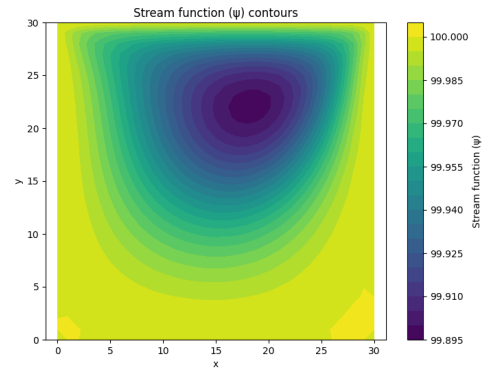


Fig. 3: Contours of the stream function illustrating the vortical structures inside the cavity.

C. Streamlines

Figure 4 provides an alternative view of the flow field using streamlines. The streamlines originate from various positions

along the cavity and clearly trace the circular motion induced by the moving lid. The central vortex is well-developed, and the streamline density near the top surface highlights the high shear due to the lid motion. The overall structure aligns well with classical solutions and reinforces the accuracy of the computed velocity field.

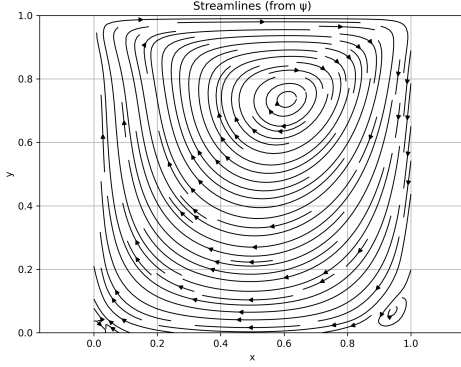


Fig. 4: Streamline plot showing the circulating flow inside the cavity.

D. Centerline Velocity Profiles

To validate the numerical results, the computed velocity profiles were compared against the benchmark data of Ghia et al. [1].

In Figure 5, the horizontal velocity component u is plotted along the vertical centerline ($x = 0.5$). The numerical results show excellent agreement with Ghia et al.'s data, especially capturing the peak velocity near the top boundary and the inflection point further down the centerline.

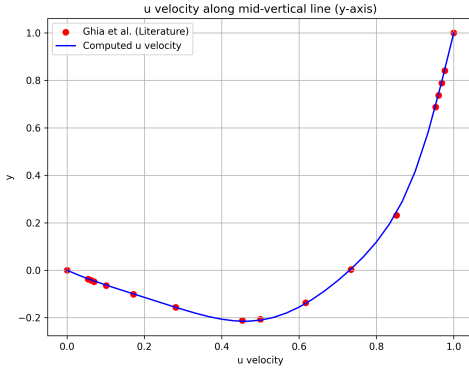


Fig. 5: Comparison of computed u -velocity along the vertical centerline with benchmark data.

Similarly, Figure 6 presents the vertical velocity component v along the horizontal centerline ($y = 0.5$). Again, the numerical solution is in close agreement with the benchmark, accurately capturing the profile's symmetric nature and the negative peak in the center due to the downward flow induced by the lid-driven circulation. The accuracy of these velocity

profiles confirms the robustness of the solver and the grid independence of the solution.

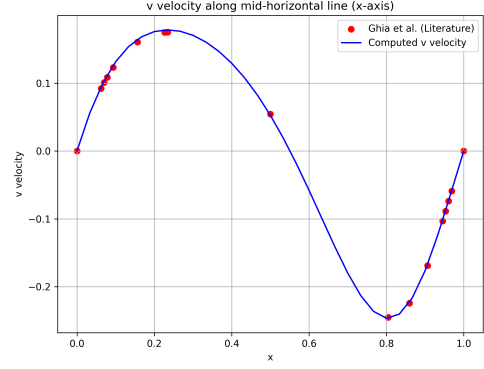


Fig. 6: Comparison of computed v -velocity along the horizontal centerline with benchmark data.

E. Summary

The results from the stream function, streamline plots, and centerline velocity profiles collectively validate the numerical scheme and its implementation. The convergence behavior shows that the chosen iterative solver is stable and efficient. The flow field characteristics are physically consistent and show good agreement with published data, establishing the reliability of the simulation even on a moderately coarse grid. These results serve as a strong foundation for extending the simulation to higher Reynolds numbers or more complex geometries.

a) *Note::* An animation of the flow field evolution is available at the following link: <https://drive.google.com/file/d/1hh2rAlWWIE3DaOoBc4zh8jD5jW0N9VHL/view?usp=sharing>

REFERENCES

- [1] U. Ghia, K. Ghia, and C. Shin, "High-re solutions for incompressible flow using the navier-stokes equations and a multigrid method," *Journal of Computational Physics*, vol. 48, no. 3, pp. 387–411, 1982, ISSN: 0021-9991. DOI: [https://doi.org/10.1016/0021-9991\(82\)90058-4](https://doi.org/10.1016/0021-9991(82)90058-4). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0021999182900584> (cit. on p. 3).